

CHAPTER

10.4

EC-06

Duration : Three Hours

Maximum Marks : 150

Q.1 to carry Q.20 one marks each and Q.21 to Q.85 carry two marks each.

1. The rank of the matrix $\begin{bmatrix} 1 & 1 & 1 \\ 1 & -1 & 0 \\ 1 & 1 & 1 \end{bmatrix}$ is

- (A) 0 (B) 1
(C) 2 (D) 3

2. $\nabla \times \nabla \times \mathbf{P}$ Where $\mathbf{P}$ is a vector, is equal to

- (A) $\mathbf{P} \times \nabla \times \mathbf{P} - \nabla^2 \mathbf{P}$ (B) $\nabla^2 \mathbf{P} + \nabla(\nabla \cdot \mathbf{P})$
(C) $\nabla^2 \mathbf{P} + \nabla \times \mathbf{P}$ (D) $\nabla(\nabla \cdot \mathbf{P}) - \nabla^2 \mathbf{P}$

3. $\iint (\nabla \times \mathbf{P}) \cdot d\mathbf{s}$ where $\mathbf{P}$ is a vector, is equal to

- (A) $\oint \mathbf{P} \cdot d\mathbf{l}$ (B) $\oint \nabla \times \nabla \times \mathbf{P} \cdot d\mathbf{l}$
(C) $\oint \nabla \times \mathbf{P} \cdot d\mathbf{l}$ (D) $\oint \nabla \cdot \mathbf{P} dv$

4. A probability density function is of the form

$$p(x) = Ke^{-\alpha|x|}, x \in (-\infty, \infty)$$

The value of K is

- (A) 0.5 (B) 1
(C) 0.5 (D) α

5. A solution for the differential equation

$$x(t) + 2x(t) = \delta(t)$$

With initial condition $x(0-) = 0$

- (A) $e^{-2t}u(t)$ (B) $e^{2t}u(t)$
(C) $e^{-t}u(t)$ (D) $e^t u(t)$

6. A low-pass filter having a frequency response $H(j\omega) = A(\omega)e^{j\phi(\omega)}$ does not produce any phase distortions if

- (A) $A(\omega) = C\omega^2, \phi(\omega) = k\omega^3$ (B) $A(\omega) = C\omega^2, \phi(\omega) = k\omega$
(C) $A(\omega) = C\omega, \phi(\omega) = k\omega^2$ (D) $A(\omega) = C, \phi(\omega) = k\omega^{-1}$

7. The values of voltage (V_D) across a tunnel-diode corresponding to peak and valley currents are V_p, V_D respectively. The range of tunnel-diode voltage for V_D which the slope of its $I - V_D$ characteristics is negative would be

- (A) $V_D < 0$ (B) $0 \leq V_D < V_p$
(C) $V_p \leq V_D < V_v$ (D) $V_D \geq V_v$

8. The concentration of minority carriers in an extrinsic semiconductor under equilibrium is

- (A) Directly proportional to the doping concentration
(B) Inversely proportional to the doping concentration
(C) Directly proportional to the intrinsic concentration
(D) Inversely proportional to the intrinsic concentration

9. Under low level injection assumption, the injected minority carrier current for an extrinsic semiconductor is essentially the

- (A) Diffusion current (B) Drift current
(C) Recombination current (D) Induced current

10. The phenomenon known as “Early Effect” in a bipolar transistor refers to a reduction of the effective base-width caused by

- (A) Electron – Hole recombination at the base
- (B) The reverse biasing of the base – collector junction
- (C) The forward biasing of emitter-base junction
- (D) The early removal of stored base charge during saturation-to-cut off switching

11. The input impedance (Z_i) and the output impedance (Z_o) of an ideal trans-conductance (voltage controlled current source) amplifier are

- (A) $Z_i = 0, Z_o = 0$
- (B) $Z_i = 0, Z_o = \infty$
- (C) $Z_i = \infty, Z_o = 0$
- (D) $Z_i = \infty, Z_o = \infty$

12. An n-channel depletion MOSFET has following two points on its $I_D - V_{GS}$ curve :

- (i) $V_{GS} = 0$ at $I_D = 12$ mA and
- (ii) $V_{GS} = -6$ Volts at $I_D = 0$ mA

Which of the following Q – point will give the highest trans – conductance gain for small signals?

- (A) $V_{GS} = -6$ Volts
- (B) $V_{GS} = -3$ Volts
- (C) $V_{GS} = 0$ Volts
- (D) $V_{GS} = 3$ Volts

13. The number of product terms in the minimized sum-of-product expression obtained through the following K – map is (where, “d” denotes don’t care states)

1	0	0	1
0	d	0	0
0	0	d	1
1	0	0	1

- (A) 2
- (B) 3
- (C) 4
- (D) 5

14. Let $x(t) \leftrightarrow X(j\omega)$ be Fourier Transform pair. The Fourier Transform of the signal $x(5t - 3)$ in terms of $X(j\omega)$ is given as

- (A) $\frac{1}{5} e^{-j\frac{3\omega}{5}} X\left(\frac{j\omega}{5}\right)$
- (B) $\frac{1}{5} e^{j\frac{3\omega}{5}} X\left(\frac{j\omega}{5}\right)$
- (C) $\frac{1}{5} e^{-j3\omega} X\left(\frac{j\omega}{5}\right)$
- (D) $\frac{1}{5} e^{j3\omega} X\left(\frac{j\omega}{5}\right)$

15. The Dirac delta function is defined as

- (A) $\delta(t) = \begin{cases} 1 & t=0 \\ 0 & \text{otherwise} \end{cases}$
- (B) $\delta(t) = \begin{cases} 1 & t=0 \\ 0 & \text{otherwise} \end{cases}$
- (C) $\delta(t) = \begin{cases} 1 & t=0 \\ 0 & \text{otherwise} \end{cases}$ and $\int_{-\infty}^{\infty} \delta(t) dt = 1$
- (D) $\delta(t) = \begin{cases} \infty & t=0 \\ 0 & \text{otherwise} \end{cases}$ and $\int_{-\infty}^{\infty} \delta(t) dt = 1$

16. If the region of convergence of $x_1[n] + x_2[n]$ is $\frac{1}{3} < |z| < \frac{2}{3}$ then the region of convergence of $x_1[n] - x_2[n]$ includes

- (A) $\frac{1}{3} < |z| < 3$
- (B) $\frac{2}{3} < |z| < 3$
- (C) $\frac{3}{2} < |z| < 3$
- (D) $\frac{1}{3} < |z| < \frac{2}{3}$

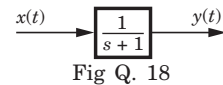
17. The open-loop function of a unity-gain feedback control system is given by

$$G(s) = \frac{K}{(s+1)(s+2)}$$

The gain margin of the system in dB is given by

- (A) 0
- (B) 1
- (C) 20
- (D)

18. In the system shown below, $x(t) = (\sin t)u(t)$ In steady-state, the response $y(t)$ will be



- (A) $\frac{1}{\sqrt{2}} \sin\left(t - \frac{\pi}{4}\right)$
- (B) $\frac{1}{\sqrt{2}} \sin\left(t + \frac{\pi}{4}\right)$
- (C) $\frac{1}{\sqrt{2}} e^{-t} \sin t$
- (D) $\sin t - \cos t$

19. The electric field of an electromagnetic wave propagation in the positive direction is given by

$$E = \hat{a}_x \sin(\omega t - \beta z) + \hat{a}_y \sin(\omega t - \beta z + \pi/2)$$

The wave is

- (A) Linearly polarized in the z -direction
- (B) Elliptically polarized
- (C) Left-hand circularly polarized
- (D) Right-hand circularly polarized

30. A two-port network is represented by ABCD, parameters given by

$$\begin{bmatrix} V_1 \\ I_1 \end{bmatrix} = \begin{bmatrix} A & B \\ C & D \end{bmatrix} \begin{bmatrix} V_2 \\ -I_2 \end{bmatrix}$$

If port – 2 is terminated by the input impedance seen at port – 1 is given by

- (A) $\frac{A + BR_L}{C + DR_L}$ (B) $\frac{AR_L + C}{BR_L + D}$
 (C) $\frac{DR_L + A}{C + BR_L}$ (D) $\frac{AR_L + B}{D + CR_L}$

31. In the two port network shown in the figure below z_{12} and z_{21} are respectively

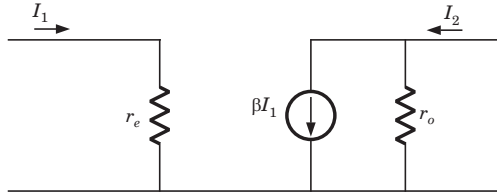


Fig Q.31

- (A) r_e and βr_o (B) 0 and $-\beta r_o$
 (C) 0 and βr_o (D) r_e and $-\beta r_o$

32. The first and the last critical frequencies (singularities) of a driving point impedance function of a passive network having two kinds of elements, are a pole and a zero respectively. The above property will be satisfied by

- (A) RL network only
 (B) RC network only
 (C) LC network only
 (D) RC as well as RL networks

33. A 2 mH inductor with some initial current can be represented as shown below, where s is the Laplace Transform variable. The value of initial current is

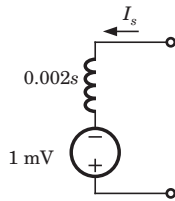


Fig Q.33

- (A) 0.5A (B) 2.0A
 (C) 1.0 A (D) 0.0 A

34. In the figures shown below, assume that all the capacitors are initially uncharged. If $v_i(t) = 10u(t)$ Volts, $v_o(t)$ is given by

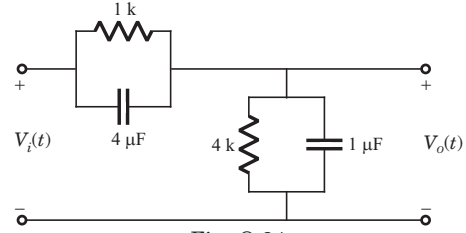


Fig. Q.34

- (A) $8e^{-0.004t}$ Volts (B) $8(1 - e^{-0.004t})$ Volts
 (C) $8u(t)$ Volts (D) 8 Volts

35. Consider two transfer functions

$$G_1(s) = \frac{1}{s^2 + as + b} \text{ And } G_2(s) = \frac{s}{s^2 + as + b}$$

The 3-dB bandwidths of their frequency responses are, respectively

- (A) $\sqrt{a^2 - 4b}$, $\sqrt{a^2 + 4b}$ (B) $\sqrt{a^2 + 4b}$, $\sqrt{a^2 - 4b}$
 (C) $\sqrt{a^2 - 4b}$, $\sqrt{a^2 - 4b}$ (D) $\sqrt{a^2 + 4b}$, $\sqrt{a^2 + 4b}$

36. A negative resistance R_{neg} is connected to a passive network N having driving point impedance $Z_1(s)$ as shown below. For $Z_2(s)$ to be positive real,

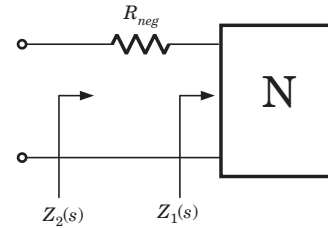


Fig Q.36

- (A) $|R_{neg}| \leq \text{Re } Z_1(j\omega)$, $\forall \omega$ (B) $|R_{neg}| \leq |Z_1(j\omega)|$, $\forall \omega$
 (C) $|R_{neg}| \leq \text{Im } Z_1(j\omega)$, $\forall \omega$ (D) $|R_{neg}| \leq \angle Z_1(j\omega)$, $\forall \omega$

37. In the circuit shown below, the switch was connected to position 1 at $t < 0$ and at $t = 0$, it is changed to position 2. Assume that the diode has zero voltage drop and a storage time t_s . For $0 < t \leq t_s$, v_R is given by (all in Volts)

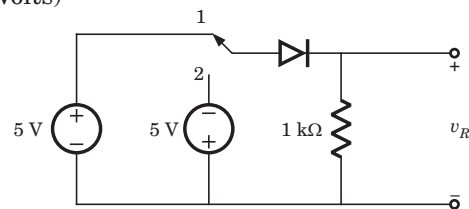


Fig Q.37

- (A) $v_R = -5$ (B) $v_R = +5$
 (C) $0 \leq v_R < 5$ (D) $-5 \leq v_R < 0$

38. The majority carriers in an n-type semiconductor have an average drift velocity v in a direction perpendicular to a uniform magnetic field $\mathbf{B}$. The electric field $\mathbf{E}$ induced due to Hall effect acts in the direction.

- (A) $\mathbf{v} \times \mathbf{B}$ (B) $\mathbf{B} \times \mathbf{v}$
 (C) along $\mathbf{v}$ (D) opposite to $\mathbf{v}$

39. Find the correct match between Group 1 and Group 2.

Group 1

Group 2

- | | |
|------------------|---------------------------------|
| E-Varactor diode | 1-Voltage reference |
| F-PIN diode | 2-High frequency switch |
| G-Zener diode | 3-Tuned circuits |
| H-Schottky diode | 4-Current controlled attenuator |
- (A) E-4, F-2, G-1, H-3 (B) E-2, F-4, G-1, H-3
 (C) E-3, F-4, G-1, H-2 (D) E-1, F-3, G-2, H-4

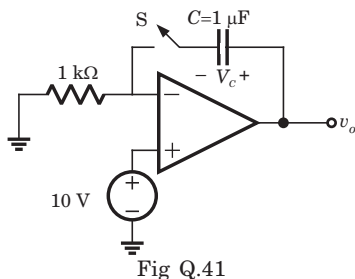
40. A heavily doped n- type semiconductor has the following data:

- | | |
|-------------------------|--|
| Hole-electron ratio | :0.4 |
| Doping concentration | : 4.2×10^8 atoms/m ³ |
| Intrinsic concentration | : 1.5×10^4 atoms/m ³ |

The ratio of conductance of the n-type semiconductor to that of the intrinsic semiconductor of same material and at same temperature is given by

- (A) 0.00005 (B) 2,000
 (C) 10,000 (D) 20,000

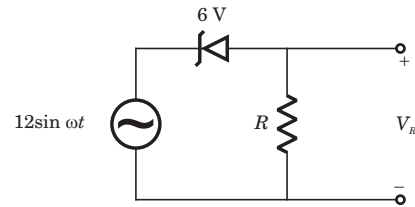
41. For the circuit shown in the following figure, the capacitor C is initially uncharged. At $t=0$ the switch S is closed. In the figures shown the OP AMP is supplied with and the ground has been shown by the symbol



The voltage V_C across the capacitor at $t=1$ is

- (A) 0 Volt (B) 6.3 Volts
 (C) 9.45 Volts (D) 10 Volts

42. For the circuit shown below, assume that the zener diode is ideal with a breakdown voltage of 6 volts. The waveform observed across R is



- (A)
- (B)
- (C)
- (D)

Q. 43 A new Binary Coded Pentary (BCP) number system is proposed in which every digit of a base-5 number is represented by its corresponding 3-bit binary code. For example, the base-5 number 24 will be represented by its BCP code 010100. In this numbering system, the BCP code 10001001101 corresponds of the following number is base-5 system

- (A) 423 (B) 1324
 (C) 2201 (D) 4231

44. An I / O peripheral device shown in Fig.(b) below is to be interfaced to an 8085 microprocessor. To select the I/O device in the I/O address range D4 H – D7 H, its chip-select ($\overline{CS}$) should be connected to the output of the decoder shown in as below:

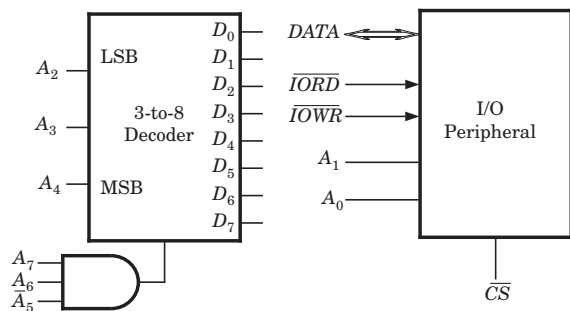


Fig Q.44

- (A) output 7 (B) output 5
(C) output 2 (D) output 0

45. For the circuit shown in figures below, two 4 – bit parallel – in serial – out shift registers loaded with the data shown are used to feed the data to a full adder. Initially, all the flip – flops are in clear state. After applying two clock pulses, the outputs of the full-adder should be

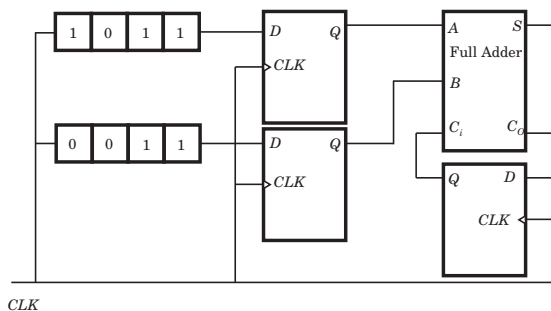


Fig Q.45

- (A) $S=0$ $C_0=0$ (B) $S=0$ $C_0=1$
(C) $S=1$ $C_0=0$ (D) $S=1$ $C_0=1$

46. A 4 – bit D / A converter is connected to a free – running 3 – big UP counter, as shown in the following figure. Which of the following waveforms will be observed at V_o ?

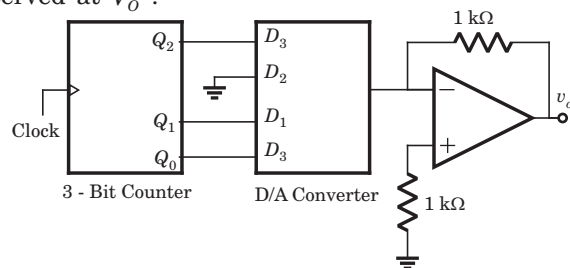
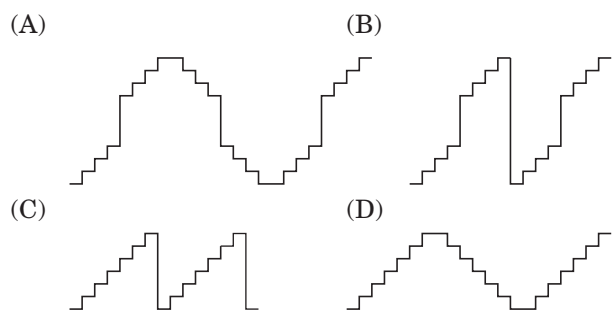


Fig Q.46



47. Two D – flip – flops, as shown below, are to be connected as a synchronous counter that goes through the following sequence

00 → 01 → 11 → 10 → 00 → ...

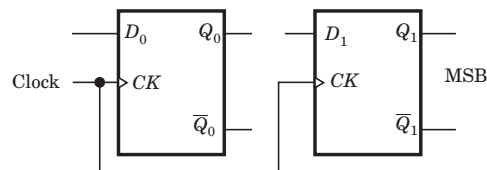


Fig Q.47

The inputs D_0 and D_1 respectively should be connected as,

- (A) $\overline{Q_1}$ and Q_0 (B) $\overline{Q_0}$ and Q_1
(C) $\overline{Q_1}Q_0$ and $\overline{Q_1}Q_0$ (D) $\overline{Q_1} \overline{Q_0}$ and Q_1Q_0

48. Following is the segment of a 8085 assembly language program

```
LXI SP, EFFF H
CALL 3000 H
:
:
:
3000 H LXI H, 3CF4
        PUSH PSW
        SPHL
        POP PSW
        RET
```

On completion of RET execution, the contents of SP is

- (A) 3CF0 H (B) 3CF8 H
(C) EFFF H (D) EFFF H

49. The point P in the following figure is stuck at 1. The output f will be

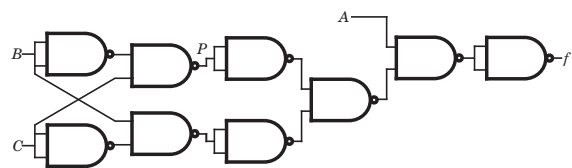


Fig Q.49

(A) $\frac{\pi}{2}$

(B) $\frac{\pi}{3}$

(C) $\frac{\pi}{4}$

(D) $\frac{\pi}{6}$

59. A linear system is described by the following state equation

$$\dot{X}(t) = AX(t) + BU(t), \quad A = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$$

The state transition matrix of the system is

(A) $\begin{bmatrix} \cos t & \sin t \\ -\sin t & \cos t \end{bmatrix}$

(B) $\begin{bmatrix} -\cos t & \sin t \\ -\sin t & -\cos t \end{bmatrix}$

(C) $\begin{bmatrix} -\cos t & -\sin t \\ -\sin t & \cos t \end{bmatrix}$

(D) $\begin{bmatrix} -\cos t & -\sin t \\ \cos t & \sin t \end{bmatrix}$

60. The minimum step- size required for a Delta – Modulator operating at 32 K , samples/sec to track the signal (here $u(t)$ is the unit function)

$$x(t) = 125t(u(t) - u(t-1)(250 - 125t)(u(t-1) - u(t-2)))$$

So that slope overload is avoided, would be

(A) 2^{-10}

(B) 2^{-8}

(C) 2^{-6}

(D) 2^{-4}

61. A zero mean white Gaussian noise is passed through an ideal lowpass filter of bandwidth 10 kHz. The output is then uniformly sampled with sampling period $t_s = 0.03$ msec. The samples so obtained would be

(A) correlated

(B) statistically independent

(C) uncorrelated

(E) orthogonal

62. A source generates three symbols with probabilities 0.25, 0.25, 0.50 at a rate of 3000 symbols per second. Assuming independent generation of symbols, the most efficient source encoder would have average bit rate as

(A) 6000 bits/sec

(B) 4500 bits/sec

(C) 3000 bits/sec

(D) 1500 bits/sec

63. The diagonal clipping in Amplitude Demodulation (using envelope detector) can be avoided if RC time – constant of the envelope detector satisfies the following condition, (here W is message bandwidth and ω_c is carrier frequency both in rad /sec)

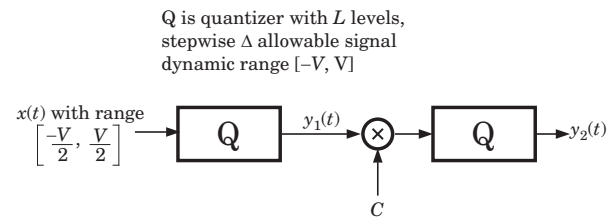
(A) $RC < \frac{1}{W}$

(B) $RC > \frac{1}{W}$

(C) $RC < \frac{1}{\omega_c}$

(D) $RC > \frac{1}{\omega_c}$

64. In the following figure the minimum value of the constant “C” , which is to be added to $y_1(t)$ and $y_2(t)$ such that $y_1(t)$ and $y_2(t)$ are different, is



(A) Δ

(B) $\frac{\Delta}{2}$

(C) $\frac{\Delta^2}{12}$

(D) $\frac{\Delta}{L}$

65. A message signal with 10 kHz bandwidth is lower side Band SSB modulated with carrier $f_{c1} = 10^6$ Hz frequency the resulting signal is then passed through a Narrow Band Frequency Modulator with carrier frequency $f_{c2} = 10^9$ Hz. The bandwidth of the output would be

(A) 4×10^4 Hz

(B) 2×10^6 Hz

(C) 2×10^9 Hz

(D) 2×10^{10} Hz

66. A medium of relative permittivity $\epsilon_{r2} = 2$ forms an interface with free – space. A point source of electromagnetic energy is located in the medium at a depth of 1 meter from the interface. Due to the total internal reflection, the transmitted beam has a circular cross-section over the interface. The area of the beam cross-section at the interface is given by

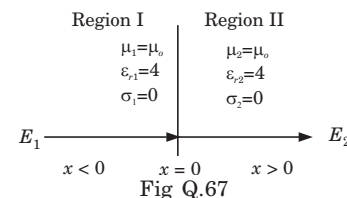
(A) $2\pi \text{ m}^2$

(B) $\pi^2 \text{ m}^2$

(C) $\frac{\pi}{2} \text{ m}^2$

(D) $\pi \text{ m}^2$

67. A medium is divide into regions I and II about $x = 0$ plane, as shown in the figure below. An electromagnetic wave with electric field $E_1 = 4\hat{a}_x + 3\hat{a}_y + 5\hat{a}_z$ is incident normally on the interface from region I. The electric file E_2 in region II at the interface is



- (A) $E_2 = E_1$ (B) $4\hat{a}_x + 0.75\hat{a}_y - 125\hat{a}_z$
 (C) $3\hat{a}_x + 3\hat{a}_y + 5\hat{a}_z$ (D) $-3\hat{a}_x + 3\hat{a}_y + 5\hat{a}_z$

68. When a plane wave traveling in free-space is incident normally on a medium having the fraction of power transmitted into the medium is given by

- (A) 8/9 (B) 1/2
 (C) 1/3 (D) 5/6

69. A rectangular wave guide having TE_{10} mode as dominant mode is having a cut off frequency 18-GHz for the mode TE_{30} . The inner broad-wall dimension of the rectangular wave guide is

- (A) 5/3 cms (B) 5 cms
 (C) 5/2 cms (D) 10 cms

70. A mast antenna consisting of a 50 meter long vertical conductor operates over a perfectly conducting ground plane. It is base-fed at a frequency of 600 kHz. The radiation resistance of the antenna in Ohms is

- (A) $\frac{2\pi^2}{5}$ (B) $\frac{\pi^2}{5}$
 (C) $\frac{4\pi^2}{5}$ (D) $20\pi^2$

Common Data for Question 71,72,73:

In the transistor amplifier circuit shown in the figure below, the transistor has the following parameters:

$$\beta_{DC} = 60, V_{BE} = 0.7 \text{ V}, h_{ie} \rightarrow \infty, h_{fe} \rightarrow \infty$$

The capacitance can be assumed to be infinite.

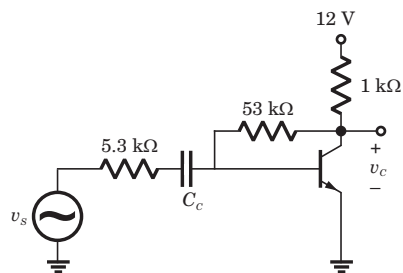


Fig Q.70

71. Under the DC conditions, the collector-to-emitter voltage drop is

- (A) 4.8 Volts (B) 5.3 Volts
 (C) 6.0 Volts (D) 6.6 Volts

72. If β_{DC} is increased by 10%, the collector-to-emitter voltage drop

- (A) increases by less than or equal to 10%
 (B) decreases by less than or equal to 10%
 (C) increases by more than 10%
 (D) decreases by more than 10%

73. The small signal gain of the amplifier v_c/v_s is

- (A) 10 (B) -5.3
 (C) 5.3 (D) 10

Common Data for Question 74, 75 :

Let $g(t) = p(t) * p(t)$ where $*$ denotes convolution and $p(t) = u(t) - u(t-1)$ with $u(t)$ being the unit step function.

74. The impulse response of filter matched to the signal $s(t) = g(t) - \delta(t-2) * g(t)$ is given as:

- (A) $s(1-t)$ (B) $-s(1-t)$
 (C) $-s(t)$ (D) $s(t)$

75. An Amplitude Modulated signal is given as

$$x_{AM} = 100(p(t) + 0.5g(t)) \cos \omega_c t$$

In the interval. One set of possible values of the modulating signal and modulation index would be

- (A) $t, 0.5$ (B) $t, 1.0$
 (C) $t, 2.0$ (D) $t^2, 0.5$

Linked Answer Question : Q.75 to Q.85 carry two marks each.

Statement of Linked Answer Question 76 & 77:

A regulated power supply, shown in figure below, has an unregulated input (UR) of 15 volts and generates a regulated output. Use the component values shown in the figure.

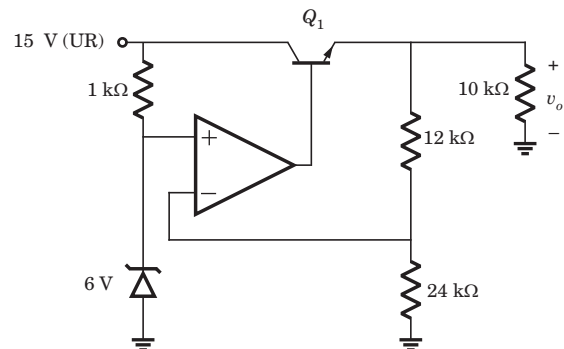


Fig Q.76

take $400\text{ }\mu\text{s}$ for an electromagnetic wave to travel from source end to load end and vice – versa. At $t = 400\text{ }\mu\text{s}$, the voltage at the load end is found to be 40 volts.

84. The load resistance is

- (A) 25 Ohms (B) 50 Ohms
(C) 75 Ohms (D) 100 Ohms

85. The steady state current through the load resistance is

- (A) 1.2 Amps (B) 0.3 Amps
(C) 0.6 Amps (D) 0.4 Amps

ANSWER

- | | | | | |
|-------|-------|-------|-------|-------|
| 1. C | 2. D | 3. A | 4. C | 5. A |
| 6. B | 7. C | 8. B | 9. A | 10. B |
| 11. D | 12. D | 13. A | 14. A | 15. D |
| 16. D | 17. D | 18. A | 19. C | 20. A |
| 21. A | 22. B | 23. D | 24. C | 25. D |
| 26. C | 27. A | 28. C | 29. A | 30. D |
| 31. B | 32. B | 33. A | 34. B | 35. B |
| 36. B | 37. D | 38. A | 39. C | 40. D |
| 41. D | 42. B | 43. D | 44. B | 45. D |
| 46. B | 47. A | 48. B | 49. D | 50. B |
| 51. C | 52. B | 53. C | 54. B | 55. D |
| 56. B | 57. C | 58. D | 59. A | 60. B |
| 61. B | 62. B | 63. D | 64. C | 65. B |
| 66. D | 67. C | 68. A | 69. C | 70. A |
| 71. C | 72. B | 73. A | 74. D | 75. A |
| 76. C | 77. B | 78. A | 79. | 80. C |
| 81. B | 82. C | 83. C | 84. | 85. |